

POKHARA UNIVERSITY

Level: Bachelor
Semester: Spring
Programme: BE
Course: Data Science and Analytics (New)

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the Knowledge Discovery from Database (KDD) process with its major phases. 7
- b) A numeric variable has the following values: 8
 32.15, 87.49, 59.60, 41.78, 12.04, 18.67, 9.31, 190.54, 101.22, 24.88. Standardize the values.
2. a) Explain power law distribution by giving real life examples. Why are power laws called heavy-tailed distribution? 8
- b) Define Pearson correlation. You are provided with the following dataset of 5 individuals, where x represents age and y represents annual salary: 7

Age	Salary in Rs
25	30000
32	45000
40	55000
48	65000
55	75000

Calculate the Pearson correlation coefficient between Age and Salary and interpret the result. Does the data suggest a strong, weak, or no linear relationship between age and salary?

3. a) What is Ordinary Least Squares (OLS) estimation, and how is the multivariate linear regression problem formulated using matrix algebra? 8

b) You are given the following data:

7

x	0.4	0.9	1.4	1.9
y	-1.8	1.2	8.5	19.1

Estimate the value of y at $x = 8$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.2$.

OR

Given a dataset with D observations on n numeric variables $x_1, \dots, x_n$, how do you compute the n principal components and determine the amount of variance explained by each of the components? Explain the process.

4. a) Consider a simple linear regression model: $Y = \beta_0 + \beta_1 X + \epsilon$, where Y is the dependent variable and X is the independent variable. Using the dataset:

8

Observation	X	Y
1	2	5
2	3	7
3	5	10
4	7	15
5	9	18

- Formulate the matrix representation of the regression model.
 - Using the estimated regression equation, predict the value of Y when $X=6$.
- b) Explain R^2 and Adjusted R^2 in the context of model evaluation in detail.
5. a) You are building a decision tree classifier. At a certain node, you have the following data:

7

8

A	B	Class label
True	False	Cat
True	True	Cat
True	True	Cat
True	False	Dog
True	True	Cat
False	False	Dog

False	False	Dog
False	False	Dog
True	True	Dog
True	False	Dog

Which attribute should the decision tree use for splitting; A or B?

Justify your answer using a purity index such as Gini impurity or entropy.

- b) Describe the CART algorithm for decision trees. How are splits determined in classification? 7
6. a) What is time series analysis? Explain its components in detail. 7

OR

How do you select the optimal lag length in ARIMA model? Explain in detail.

- b) What conditions must be satisfied to conclude that event A causes event B? Describe. Why is Granger causality not considered true causality? 8
7. Write short notes on: (Any two) 2×5
- a) Statistical significance
- b) PCA
- c) Causation vs. correlation

Exam	Final Internal Examination 2025 Fall		
Level	B E	F M	100
Program	Computer	PM	45
Year/ Part	IV/I	Time	3 Hrs

Subject: Data Science and Analytics

*Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary
Attempt all the questions.*

1.	a)	How would you define data science? Explain the KDD process in brief.	7															
	b)	A numeric variable has the following values. 46.93, 138.07, 64.77, 51.46, 15.38, 15.37, 11.32, 223.21, 120.23, 19.85, 19.80, 12.39, 55.79, 86.31, 3.99, 101.51, 30.71, 94.28, 102.39, 18.04 Standardize the values.	8															
2.	a)	Based on a survey, the variable x takes the following values. 0.76, 1.53, 0.12, 2.44, 2.66, 3.84, 1.03, 6.31, 2.03, 4.61, 9.53, 10.39, 3.69, 6.63, 6.4, 5.01, 2.99, 1.18, 0.35, 0.44, 1.99, 7.04, 3.05, 7.89, 1.44, 1.11, 7.53, 10.4, 1.72, 4.96. Do you think x follows a normal or an exponential type distribution? Support your answer with appropriate plots.	8															
	b)	Differentiate discrete and continuous numeric variables with examples. What are nominal and ordinal categorical variables? Explain with examples	7															
3.	a)	Explain the importance of Pearson's correlation coefficient . Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula: X = 10, 13, 15, 17, 19 and Y = 5, 10, 15, 20, 25.	7															
	b)	The following table presents a hypothetical cross-tabulation between 'food preferences' and 'gender'. <table style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Food Preference</th><th>Female</th><th>Male</th><th>Other</th></tr> </thead> <tbody> <tr> <td>Likes buff momo</td><td>8</td><td>75</td><td>1</td></tr> <tr> <td>Likes chicken momo</td><td>60</td><td>3</td><td>0</td></tr> <tr> <td>Likes veg momo</td><td>23</td><td>19</td><td>3</td></tr> </tbody> </table> Do you think food preferences are correlated with gender? Justify using chi squared test with significance level 0.05. The critical value for the Chi-Squared statistic with 4 degrees of freedom at a significance level of 0.05 is 9.4877.	Food Preference	Female	Male	Other	Likes buff momo	8	75	1	Likes chicken momo	60	3	0	Likes veg momo	23	19	3
Food Preference	Female	Male	Other															
Likes buff momo	8	75	1															
Likes chicken momo	60	3	0															
Likes veg momo	23	19	3															
4.	a)	What is multicollinearity in regression analysis? How do variance-inflation factors (VIF) help detect it, and what steps can be taken if high multicollinearity is found?	7															
	b)	Compare and contrast Principal Component Analysis (PCA) and K-means clustering, explaining their mathematical foundations and practical applications.	8															

5.	a) You are given the following data. Assuming that there is an unknown functional relationship $y = f(x)$, estimate the value of y at $x = 2.9$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.5$. <table border="1"><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>-2.3</td><td>0.5</td><td>9.3</td><td>20.8</td></tr></table> b) What is the role of R-squared and Adjusted R-squared? Compute R-squared for a regression model to measure how well it explains the variation in the dependent variable Y. <table border="1"><tr><td>Hours Studies (X)</td><td>Exam Score (Y)</td></tr><tr><td>1</td><td>50</td></tr><tr><td>2</td><td>60</td></tr><tr><td>3</td><td>70</td></tr><tr><td>4</td><td>80</td></tr></table>	x	1	2	3	4	y	-2.3	0.5	9.3	20.8	Hours Studies (X)	Exam Score (Y)	1	50	2	60	3	70	4	80	8
x	1	2	3	4																		
y	-2.3	0.5	9.3	20.8																		
Hours Studies (X)	Exam Score (Y)																					
1	50																					
2	60																					
3	70																					
4	80																					
6.	a) Explain the concepts of Decision Trees and Logistic Regression in machine learning. b) Explain about autocorrelation and stationarity. The following time series represents yearly production data: <table border="1"><tr><td>t</td><td>X_t</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>6</td></tr><tr><td>3</td><td>8</td></tr><tr><td>4</td><td>10</td></tr><tr><td>5</td><td>12</td></tr><tr><td>6</td><td>14</td></tr></table> (a) Compute autocorrelation at lag $k=1$ (b) Divide the data into two equal parts and compute the mean (c) Comment on stationarity	t	X_t	1	4	2	6	3	8	4	10	5	12	6	14	7						
t	X_t																					
1	4																					
2	6																					
3	8																					
4	10																					
5	12																					
6	14																					
7.	Write short notes on: (Any two) a) OLS estimation b) Causation vs. correlation c) Zipf's law	2×5																				

*** Best of Luck ***

Course: CMP 360 – Data Science and Analytics

Time: 3 Hours

Full Marks: 100

Semester: 7th computer

1. a) Compare structured and unstructured data. Discuss challenges in handling unstructured data. [7]
b) Explain the complete Knowledge Discovery in Databases (KDD) process. Illustrate each step using an example. [8]
2. a) compute Karl Pearson's correlation coefficient and interpret the result. -

X	4	7	8	9	5	3
Y	7	9	5	10	8	5

- b) Discuss histogram, binomial distribution, and Zipf's law with appropriate examples.
3. a) A survey recorded the following information about purchasing habit according to gender. Perform chi-square test for independent of attributes at 5 % level of significance. [7]

	Purchase	Not purchase
Male	35	25
Female	40	50

- b) Explain: [8]

R^2 and Adjusted R^2

Multicollinearity

Variance Inflation Factor (VIF)

Discuss how multicollinearity affects interpretation of regression coefficients.

4. a) Derive the Nadaraya–Watson kernel regression estimator and explain the role of kernel function and bandwidth.[7]

b) Fit multiple linear regression model.[8]

Study time in hours	2	4	6
Score on math	70	80	90
Score on reading	65	75	85

5. a) Explain the CART algorithm for classification. Discuss splitting criteria.[7]

b) Given the covariance matrix: [8]

$$\Sigma = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

- i) Find the eigenvalues.
ii) Determine the proportion of total variance explained by the first principal component.
6. a) Define stationarity and autocorrelation. Explain ARMA and ARIMA models with suitable mathematical formulation.[7]
b) Given the estimated logistic regression model:[8]

$$\log\left(\frac{p}{1-p}\right) = -2 + 0.8X$$

Compute the probability when $X = 3$. Also, differentiate logistic regression model with linear regression model.

7. Write short note. Any two.[10]

- Memory less property
- Causation vs correlation
- Causal Directed Acyclic Graph (DAG)



Pokhara University
Everest Engineering College
Final Internal Assessment
Fall-2025

Level: Bachelor

F.M. 100

Program: BE CMP(7th Sem)

P.M. 45

Section: A/B

Time: 3hrs

Subject: Data Science and Analytics

Attempt all the questions.

- 1 a) Explain different data cleaning process. 7
 b) Explain Knowledge Discovery in databases. A numeric variable has the following values. 21.5, 67.8, 42.7, 8.3, 102.55, 57.89, 22.77. Standardize the values. 8
- 2 a) Explain and interpret Zipf's law. Also list application and limitation of Zipf's law. 7

Age (x)	Salary (y, in \$1,000s)
25	30
32	45
40	55
48	65
55	75

8

Calculate and interpret Pearson's co-relation coefficient. Explain the purpose of T-value, P value and ANOVA test.

- 3 a) For the given dataset, find the regression equation using matrix approach. 8

Observation	x_1	x_2	y
1	1	2	6
2	2	1	8
3	3	3	12

7

- b) Explain R^2 and adjusted R^2 in the context of model evaluation in detail.

OR

Derive the equation of Nadaraya-Watson kernel regression estimator.

4 a) Given data = {2,3,4,5,6,7;1,5,3,6,7,8}. Compute the principal component using PCA Algorithm.

b) Given a new dataset to demonstrate non-parametric regression:

$$X = [1, 2, 3, 4, 5]$$

$$Y = [2, 4, 1, 5, 3]$$

Estimate $\hat{y}$ at $x = 2.5$ using the Nadaraya-Watson kernel regression with Gaussian kernel:

$$\hat{y}(x) = \frac{\sum_{i=1}^n K_h(x - X_i) Y_i}{\sum_{i=1}^n K_h(x - X_i)}$$

where $K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$ and bandwidth $h = 1$.

5 a) Describe CART algorithm for decision trees. How are splits determined in classification?

b) Make a decision tree from the following dataset. Assume necessary data and algorithm.

Outlook	Temperature	Humidity	Windy	PlayGolf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

$\frac{1}{2} \times \frac{1}{6}$

$5 \times \frac{2}{5}$

OR

Cluster the following eight 2D points using the K Means algorithm with $K=3$ and perform 2 iterations.

Point	Coordinates
A_1	(2,10)
A_2	(2,5)
A_3	(8,4)
A_4	(5,8)
A_5	(7,5)
A_6	(6,4)
A_7	(1,2)
A_8	(4,9)

- 6 a) Explain the components of time series analysis in details. 7
b) Briefly describe ARMA and ARIMA models. How do you select the optimal lag length in ARIMA model? 8
- 7 Write short notes on: (Any two) 2*5=10
a) Cross tabulation
b) Logistic Regression
c) Causation Vs Correlation

****Good Luck****

NEPAL ENGINEERING COLLEGE

Level: Bachelor

Programme: BE

Course: Data Science and Analytics.

Year : 2026

Full Marks: 100

Pass Marks: 45

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. (a) Explain the Knowledge Discovery process from data (KDD) with its major phases. How does data integration address the challenge of combining the data from multiple sources, multiple formats and forms? 8
(b) Differentiate between Structured and Unstructured Data and Distinguish Between Numeric and Categorical Data. 7
2. (a) Explain Pearson correlation and how it measures the relationship between two numeric variables? Show it by considering an example. 8
(b) Describe the Empirical Distribution of Numeric Data. Explain Histogram, Normal Distribution, Exponential Distribution and Power Law Distribution with Examples. 7
- 3 (a) Explain Multivariate Linear Regression with its Matrix Formulation. Explain R squared and R squared adjusted with their formula. 8
(b) You are given the following two-dimensional datasets consisting of 6 observations: (2,1), (3,5), (4,3), (5,6), (6,7), (7,8). Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset. 7
4. (a) You are given the following data. 8

x	1	2	3	4
y	-2.3	0.5	9.3	20.8

Assuming that there is an unknown functional relationship $y = f(x)$, estimate the value of y at $x = 2.9$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.5$.

- (b) What is the importance of bandwidth in kernel regression? 7
How is it selected? Explain with Example.
5. (a) Describe CART algorithm for decision tree with example. 8
Explain how Logistic Regression is applied in classification Problem.
- (b) Apply the K-means Algorithm with $K=2$ on the following Dataset: (160,55), (175,65), (172,60), (180,70), (178,68), 7
(165,58).
6. (a) What is time series analysis? Explain its components in 8
detail.

OR

What conditions must be satisfied to conclude that event A causes event B? Describe why granger causality is not considered a true causality.

- (b) Define ARMA and ARIMA models. What is the role of AIC 7
and BIC in model selection?
7. Write Short Notes on (Any two):
- (a) Permutation tests for Variable Importance
 - (b) Zipf's Law
 - (c) Multi-collinearity and variance-inflation factors (VIF).

[10]

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor

Semester : VII_Fall

Year : 2025

Program: B.E. Computer

F.M. : 100

Course: Data Science and Analytics

P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Explain the Knowledge Discovery from Database (KDD) process in detail. How does it relate to the broader Data Science Pipeline? (7)
b) Distinguish between structured and unstructured data with appropriate examples. Why is data integration a challenging step in preprocessing data from multiple sources? (8)
2. a) Compare Numeric and Categorical variables. Explain how the Pearson correlation coefficient is used to measure the strength of a relationship between two numeric variables. (7)
b) Define a Power Law distribution and explain why it is referred to as a "heavy-tailed" distribution. How does it differ from a Normal distribution in real-world data? (8)
3. a) Describe the Multivariate Linear Regression model. Show the matrix formulation for the model and explain how the Ordinary Least Squares (OLS) estimation is used to find the coefficients. (8)
b) A researcher uses two predictors to model a target variable. Explain the significance of R-squared and Adjusted R-squared. Why is Adjusted R-squared preferred when comparing models with different numbers of predictors? (7)
4. a) Derive the Nadaraya-Watson kernel regression estimator. Discuss the critical role of the bandwidth parameter in determining the performance of this non-parametric model. (8)

OR

You are given the following dataset

x	0.5	1.0	1.5	2.0
y	-2.3	0.5	9.3	20.8

Estimate the value of y at $x = 1.3$ using the Nadaraya-Watson regression estimator with a bandwidth $h = 0.2$. (8)

b) Explain Principal Component Analysis (PCA) as a tool for identifying latent variables. Briefly explain the steps involved in it. (7)

5. a) Describe the CART (Classification and Regression Tree) algorithm. Explain the difference between Information Gain and the Gini Index as heuristics for attribute selection during tree induction. (7)

b) Illustrate the steps of the k-means clustering algorithm. How do we determine if the algorithm has reached convergence, and what are the limitations of using Euclidean distance for mixed-type data? (8)

OR

Apply the K-Means algorithm with $K=2$ on the following dataset: (160, 55), (175, 65), (172, 60), (180, 70), (178, 68), (165, 58). Use the first two data points (160, 55) and (175, 65) as the initial centroids. Perform the clustering for two iterations and show the cluster membership after each step. (8)

6. a) Discuss Logistic Regression and how it models the probability of a binary outcome. Why is it often preferred over linear regression for classification tasks like species habitat prediction? (7)

b) Define Autocorrelation and Stationarity in the context of time-series analysis. List and explain the four components of a time series: trend, seasonal, cyclical, and irregular variations. (8)

OR

Compare ARMA and ARIMA models. Explain why the "Integrated" (I) component is necessary when dealing with non-stationary data. (8)

7) Write short notes on (Any two): ($2 \times 5 = 10$)

a) Granger Causality

b) Statistical Significance and p-value

c) Akaike (AIC) and Bayesian (BIC) Information Criteria